STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-AXIOM-C

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2689-2698 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 2700-2740, the complete author proof (39 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 28 of queue_sdga.json (pb=933, stmt 2689-2698, proof 2700-2740).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 09QJ: sources/stacks-sdga/pinned-original-source.tex, source lines 2740–2740. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. The differential graded category $Mod^{dg}(\mathcal{A}, d)$ satisfies axiom (C) formulated in Differential Graded Algebra, Situation 09QJ.*

Proof. Let $f : \mathcal{K} \to \mathcal{L}$ be a homomorphism of differential graded $\mathcal{A}$-modules. By the above we have an admissible short exact sequence

$$0 \to \mathcal{L} \to C(f) \to \mathcal{K}[1] \to 0$$

To finish the proof we have to show that the boundary map

$$\delta : \mathcal{K}[1] \to \mathcal{L}[1]$$

associated to this (see discussion above) is equal to $f[1]$. For the section $s : \mathcal{K}[1] \to C(f)$ we use in degree n the embedding $\mathcal{K}^{n+1} \to C(f)^n$. Then in degree n the map π is given by the projections $C(f)^n \to \mathcal{L}^n$. Then finally we have to compute

$$\delta = \pi \circ \mathrm{d}_{C(f)} \circ s$$

(see discussion above). In matrix notation this is equal to

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \mathrm{d}_{\mathcal{L}} & f \\ 0 & -\mathrm{d}_{\mathcal{K}} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = f$$

as desired. □

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.